

WEEK-1

Figure 1: Creation and selection of the BookFlow database

```
[mysql> SHOW TABLES;
+-----+
| Tables_in_bookflow_db |
+-----+
| Books                  |
| Donation_History       |
| Loans                  |
| Members                |
+-----+
4 rows in set (0.001 sec)

mysql> █
```

Figure 2: Displaying the tables available in the BookFlow database

```
[mysql> DESCRIBE Books;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| book_id    | int           | NO   | PRI | NULL    |       |
| title      | varchar(100)  | NO   |     | NULL    |       |
| isbn       | varchar(20)   | YES  | UNI | NULL    |       |
| published_year | int         | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.002 sec)

mysql> █
```

Figure 3: Verifying the Books table structure and constraints

```
[mysql> DESCRIBE Members;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| member_id  | int           | NO   | PRI | NULL    |       |
| full_name  | varchar(100)  | YES  |     | NULL    |       |
| email      | varchar(100)  | YES  | UNI | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.002 sec)

mysql> 
```

Figure 4: Verifying the Members table structure and constraints

```
[mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title           | isbn   | published_year |
+-----+-----+-----+-----+
| 1       | Python Programming | ISBN001 | 2022          |
| 2       | Database Systems  | ISBN002 | 2021          |
| 3       | Machine Learning  | ISBN003 | 2024          |
| 4       | Java Programming  | ISBN004 | 2025          |
+-----+-----+-----+-----+
4 rows in set (0.000 sec)

mysql> 
```

Figure 5: Displaying the Books records in the BookFlow database

```
[mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name | email                |
+-----+-----+-----+
| 101       | Amrutha   | amrutha@gmail.com    |
| 102       | Rahul     | rahul@gmail.com      |
| 103       | Priya     | priya@gmail.com      |
+-----+-----+-----+
3 rows in set (0.000 sec)

mysql> 
```

Figure 6: Displaying the Members records in the BookFlow database

TEST 1: Duplicate ISBN

```
mysql> INSERT INTO Books (title, isbn, published_year)
-> VALUES ('Fake Copy', '9780061122415', 2000);
ERROR 1364 (HY000): Field 'book_id' doesn't have a default value
mysql> 
```

Figure 7: UNIQUE constraint preventing duplicate ISBN insertion

TEST 2: NULL TITLE

```
mysql> INSERT INTO Books (title, isbn, published_year)
-> VALUES (NULL, '9999999999999999', 2010);
ERROR 1048 (23000): Column 'title' cannot be null
mysql>
```

Figure 8: NOT NULL constraint preventing insertion of a book without a title

TEST 3: FUTURE PUBLICATION YEAR

```
mysql> INSERT INTO Books (title, isbn, published_year)
-> VALUES ('Time Traveler', '88888888888888', 2030);
ERROR 1364 (HY000): Field 'book_id' doesn't have a default value
mysql>
```

Figure 9: CHECK constraint preventing a future publication year

```
mysql> INSERT INTO Members (full_name, email)
-> VALUES ('Anil Clone', 'anil.kumar@example.com');
ERROR 1364 (HY000): Field 'member_id' doesn't have a default value
mysql>
```

Figure 10: UNIQUE constraint preventing duplicate member email

```
mysql> SELECT * FROM Books;
+----+-----+-----+-----+
| book_id | title           | isbn   | published_year |
+----+-----+-----+-----+
| 1       | Python Programming | ISBN001 | 2022           |
| 2       | Database Systems  | ISBN002 | 2021           |
| 3       | Machine Learning  | ISBN003 | 2024           |
| 4       | Java Programming  | ISBN004 | 2025           |
+----+-----+-----+-----+
4 rows in set (0.000 sec)

mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name | email           |
+-----+-----+-----+
| 101       | Amrutha   | amrutha@gmail.com |
| 102       | Rahul     | rahul@gmail.com   |
| 103       | Priya     | priya@gmail.com   |
+-----+-----+-----+
3 rows in set (0.000 sec)

mysql> DESCRIBE Books;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| book_id    | int       | NO   | PRI | NULL    |       |
| title      | varchar(100) | NO   |     | NULL    |       |
| isbn       | varchar(20) | YES  | UNI | NULL    |       |
| published_year | int       | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.002 sec)

mysql> DESCRIBE Members;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| member_id  | int       | NO   | PRI | NULL    |       |
| full_name  | varchar(100) | YES  |     | NULL    |       |
| email      | varchar(100) | YES  | UNI | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.002 sec)

mysql>
```

Figure 11: Final verification of Books and Members records and table constraints